

Baoren Liu

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

Bachelor of Science in Mathematics & Computer Science; GPA: 3.92

Expected: May 2028

Relevant Courseworks: Data Structures & Algorithms, Database Systems, Linear Algebra, Probability & Statistics

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, HTML, CSS, JavaScript, TypeScript, Kotlin, Bash

Technologies: React, Next.js, Flask, PyTorch, LangChain, NumPy, Pandas, scikit-learn, ChromaDB, Hugging Face

Tools: Git, GitHub, CI/CD, PostgreSQL, Vercel, Docker, Google Cloud Platform, Microsoft Azure, Linux, Ollama, Isaac Sim, Codex

EXPERIENCE

University of Illinois Urbana-Champaign

July 2026 – Present

Undergraduate Researcher

Champaign, IL

- Developing an **agentic** benchmark using **Isaac Sim** for robots to actively diagnose and recover from task failures
- Implementing configurable simulation environments, scripted diagnostic actions, recovery state machines, and automated episode evaluation in Python

Mass General Brigham

June 2024 – Present

AI Research Intern

Boston, MA

- Built an end-to-end retrieval-augmented generation pipeline to process, embed, retrieve, and classify evidence across **115M+ proprietary clinical records** in a HIPAA-compliant environment
- Automated evaluation across retrieval methods, embedding models, reranking, LLMs, and traditional classifiers using cross-validation and bootstrapping
- Co-authored** a research paper on the system, which achieved a **0.927 F1-score** for dementia identification and is currently under peer review
- Automated deployment of the project website using **GitLab CI/CD**, SSH-managed Linux runners, and webhooks, eliminating manual deployment steps

PROJECTS

RideList | *Next.js, Mantine, PostgreSQL, Cloudflare API, Vercel, Google Cloud, Azure, Git* January 2026 – Present

- Built and launched a bilingual campus ridesharing app with Next.js, supporting **60+ trips, 75+ messages**, and **7 completed bookings**
- Designed a scalable PostgreSQL schema and optimized queries for ride discovery, messaging, and trip management
- Secured communication between Vercel-hosted APIs and a self-hosted database using a custom **certificate authority and mTLS**
- Built privacy-preserving geospatial ride matching with **PostGIS** and Google Maps APIs
- Implemented real-time messaging, trip synchronization, and automated PostgreSQL backups using Firebase and Bash scripts on Linux

RateMySchedule | *React, Flask, PostgreSQL, Docker, Ollama, Git*

March 2026 – May 2026

- Collaborated in a **four-person Git workflow** to build a full-stack tool for importing schedules and evaluating semester difficulty
- Architected a layered Flask API with OpenAPI documentation, PostgreSQL schema design, and Docker Compose
- Reverse-engineered** UIUC course-catalog XML requests, built a resilient data-scraping pipeline, and integrated Ollama for schedule analysis

LHS Schedule | *React, Material UI, Firebase, Netlify*

July 2020 – December 2024

- Built a scheduling app with real-time schedule updates, cross-device synchronization, offline access, PDF schedule importing, and social features
- Supported **1,400+ registered users and 900 daily active users** by parallelizing API requests, caching data with a **service worker**, and deploying on Netlify

AWARDS

6th of 131 teams overall; 1st in Beginner Division, UIUC SIGPwny Fall CTF Competition – 2025